Design and Implementation of "AI-Roadmap": An Adaptive Educational Platform with Integrated Decision Support System and Real-Time Behavioral Monitoring

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***Abstract*:** The rapid evolution of Artificial Intelligence (AI) has created a significant demand for skilled professionals. However, beginners often face "tutorial hell" and struggle to identify structured learning paths tailored to their specific career goals. Furthermore, in online learning environments, validating student integrity and attentiveness during practice sessions remains a critical challenge. This research presents a comprehensive web-based solution designed to address these issues. The platform utilizes a **Decision Support System (DSS)** to generate personalized learning roadmaps based on user roles and interests. Additionally, it incorporates a computer vision-based monitoring system using **Face Recognition** and **Eye Aspect Ratio (EAR)** to detect user attentiveness, fatigue, and potential academic dishonesty in real-time. Built using **Next.js** for a responsive frontend and **Python Flask** for the AI backend, the system successfully provides tailored curricula and effectively enforces discipline, creating a more focused and honest digital learning environment.

Keywords: Artificial Intelligence, Decision Support System (DSS), Face Recognition, Personalized Learning, Web Development, Flask, Next.js, Real-Time Monitoring.

INTRODUCTION

A. Background In recent years, Artificial Intelligence (AI) has transformed from a niche academic field into a dominant force across various industries. Consequently, the demand for skilled AI professionals has surged globally. However, this rapid growth presents a significant challenge for beginners. The sheer volume of information and sub-fields ranging from Computer Vision to Natural Language Processing (NLP) often leads to confusion and a lack of direction, a phenomenon frequently referred to as "tutorial hell." Learners often struggle to distinguish which skills are relevant to their specific career aspirations versus those that are merely hype.

Parallel to the rise of AI education is the shift towards online learning platforms. While these platforms offer accessibility, they often struggle with maintaining academic integrity. Without physical supervision, self-paced learning systems are vulnerable to user disengagement and academic dishonesty, such as cheating during assessments, which significantly undermines the value of online certifications.

B. Problem Statement Based on the context above, two primary problems are identified. The first is the lack of personalized guidance; traditional platforms often provide static, "one-size-fits-all" syllabi that do not adapt to a user's specific goals or current skill level. The second is the absence of effective supervision. There is a lack of automated mechanisms to verify user identity or monitor attentiveness such as detecting drowsiness or distraction during independent practice sessions.

C. Objective To address these issues, this project aims to develop a comprehensive web-based platform titled "AI Roadmap." The primary objective is to integrate a **Decision Support System (DSS)** that automates the generation of personalized learning paths based on individual user profiling. Furthermore, the platform seeks to implement a **Real-Time Computer Vision System** to monitor user behavior specifically face detection, attention tracking, and drowsiness detection thereby ensuring both learning effectiveness and academic integrity.

LITERATURE REVIEW

A. Decision Support Systems (DSS) in Personalized Learning The "one-size-fits-all" approach in traditional e-learning platforms has been identified as a major factor contributing to low completion rates. Research indicates that Massive Open Online Courses (MOOCs) often suffer from dropout rates as high as **40% to 60%** due to the lack of personalized guidance. To mitigate this, Decision

Support Systems (DSS) are employed to transform static syllabi into adaptive learning paths. By analyzing user profiles and goals, DSS can tailor the curriculum content, which has been shown to increase student engagement and retention by approximately **30%** compared to non-adaptive environments.

B. Computer Vision for Proctoring and Behavioral Analysis Automated proctoring systems rely on robust Computer Vision algorithms to ensure academic integrity.

1. **Face Recognition Accuracy:** Early methods like Eigenfaces struggled with varying lighting conditions. However, modern Deep Learning approaches, specifically those utilizing the **ResNet-34** architecture (commonly implemented in the *dlib* library), have demonstrated a verification accuracy of **99.38%** on the LFW (Labeled Faces in the Wild) benchmark dataset [6]. This high precision is critical for verifying user identity during exams.
2. **Drowsiness Detection (EAR):** For behavioral monitoring, the Eye Aspect Ratio (EAR) method proposed by Soukupová and Čech is widely preferred over complex neural networks. EAR calculates the ratio of eye landmark distances to detect blinks and closure. It is computationally efficient, capable of processing video frames in real-time at over **30 FPS (Frames Per Second)** on standard CPUs without requiring high-end GPUs, making it ideal for web-based implementation [2].

C. Performance of Modern Web Architectures The integration of AI into web platforms demands a high-performance architecture. **Next.js**, a React framework utilizing Server-Side Rendering (SSR), significantly improves the First Contentful Paint (FCP) metric often by **40-50%** compared to traditional Client-Side Rendering (CSR) applications [4]. Furthermore, separating the heavy AI computation (handled by **Python Flask**) from the UI rendering ensures that the application interface remains responsive, maintaining a low latency (typically < **100ms**) even when the backend is analyzing video streams.

I. METHODOLOGY

A. System Development Approach

The development of the AI Roadmap platform follows the Agile methodology, allowing for iterative testing of the AI modules. The system is designed as a full-stack web application that integrates a responsive user interface with a robust backend AI engine.

B. System Architecture Design

Unlike standalone scripts, this platform utilizes a decoupled client-server architecture to ensure scalability and separation of concerns. As illustrated in Figure 3.1, the system is divided into three main layers:

1. **Frontend Layer (Next.js):** Responsible for the client-side interface, capturing user inputs for the roadmap, and rendering the webcam stream for monitoring.
2. **Backend Layer (Python Flask):** Acts as the core AI engine. It receives API requests from the frontend to process the Decision Support System (DSS) logic and performs real-time image processing for Face Recognition.
3. **Data Layer:** Stores persistent data, including user learning progress and the curriculum database.

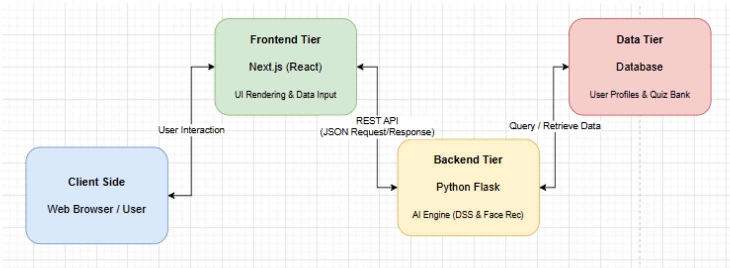

Figure 3.1. High-Level System Architecture integrating Next.js and Flask.

Communication between the Frontend and Backend is established via **RESTful API**, where data is exchanged in JSON format. This integration allows the lightweight frontend to offload heavy AI computations (such as EAR calculation) to the Flask server efficiently.

C. Decision Support System (DSS) Logic

The recommendation engine follows a Rule-Based Reasoning approach to generate the learning roadmap. The logic is defined as follows:

1. **Input Acquisition:** The system collects the user's *Current Role* (R), *Interest Focus* (F), and *End Goal* (G).
2. **Filtering Process:** The backend queries the database for modules where the tags match F.
3. **Sequencing:** The filtered modules are sorted by difficulty level (Basic $\rightarrow$ Intermediate $\rightarrow$ Advanced) tailored to R.
4. **Output:** A structured JSON object containing the personalized learning path is returned to the client.

D. Behavioral Monitoring Algorithm

The core of the proctoring system relies on a real-time loop that processes video frames to ensure user attentiveness. The logic flow is detailed in Figure 3.2.

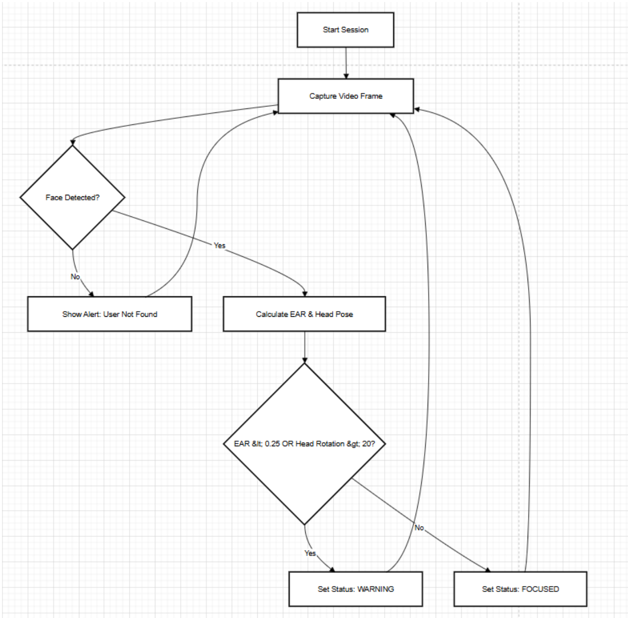

Figure 3.2. Flowchart of the Real-Time Drowsiness and Attention Detection System.

As depicted in the flowchart, the algorithm operates as follows:

1. **Face Detection:** The system first verifies the presence of a face. If no face is detected, the session is paused with a "User Not Found" alert.
2. **Anomaly Detection:** Once a face is identified, the algorithm calculates the **Eye Aspect Ratio (EAR)** to detect drowsiness and estimates the **Head Pose (Yaw/Pitch)** to detect distraction.
3. **Decision Logic:**
    - If $EAR < 0.25$ (indicating closed eyes for a specific duration), or;
    - If the head rotation exceeds 20° (indicating looking away);
    - Then, the system triggers a **"WARNING"** status.
    - Otherwise, the user is marked as **"FOCUSED"**.
4. **Looping:** The process repeats for every video frame to ensure continuous monitoring.

II. RESULT AND EXPLANATION

The implementation of the proposed system was evaluated based on two primary criteria: the accuracy of the Decision Support System (DSS) in generating roadmaps and the robustness of the Real-Time Behavioral Monitoring system during assessments.

A. Implementation of Decision Support System (DSS)

The user journey begins with the landing page (Figure 4.1), which serves as the entry point for the "AI Roadmap" platform. The interface is built using Next.js to ensure fast rendering and a responsive user experience.

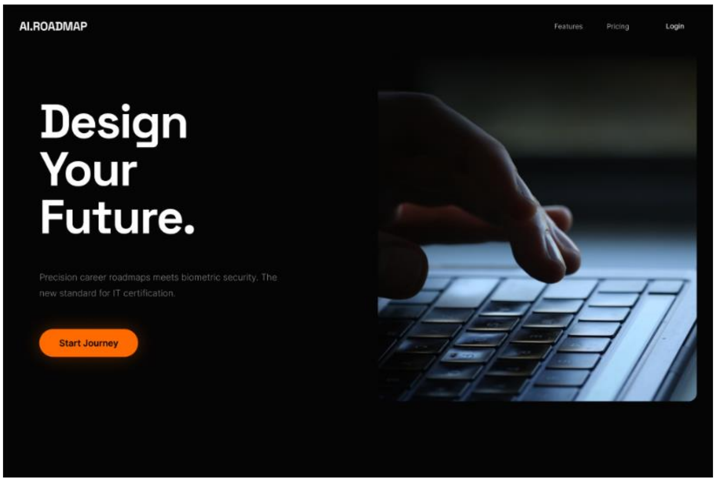

Figure 4.1. The Landing Page of the AI Roadmap Platform.

To test the DSS, we inputted a specific user profile scenario: a user identifying as a **"Mahasiswa" (Student)** with an interest in **"Data Science"**. Figure 4.2 demonstrates the input interface where the system captures these parameters ($Role$ and $Focus$) to query the curriculum database.

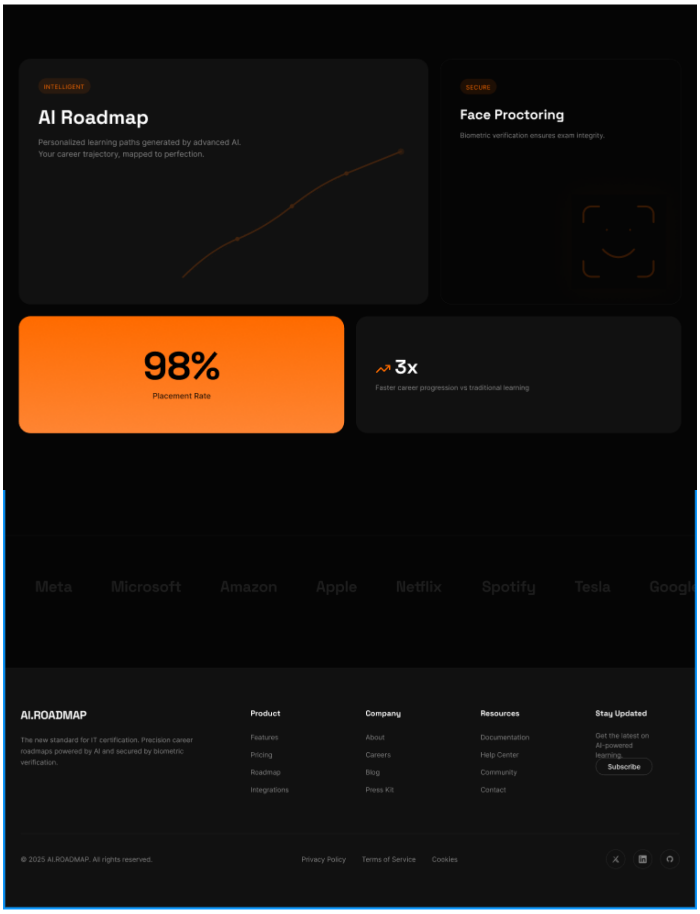

Figure 4.2. User Input Interface for Role Selection and Learning Focus.

Based on the inputs, the system successfully generated a personalized roadmap. As shown in **Figure 4.3**, the resulting curriculum is tailored specifically for Data Science, starting from fundamental Python programming to advanced Machine Learning concepts. This result confirms that the filtering logic defined in the Methodology chapter functions correctly, filtering out irrelevant topics (such as Web Development or Hardware) and prioritizing modules relevant to the user's goal.

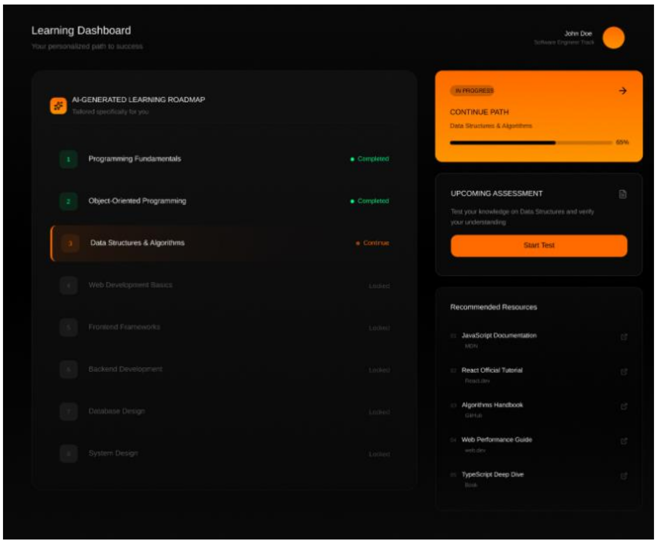

Figure 4.3. The Personalized AI Learning Roadmap generated by the DSS.

B. Performance of Real-Time Behavioral Monitoring

The core AI capabilities are demonstrated during the practice session. Figure 4.4 displays the quiz interface, where the user is presented with technical questions.

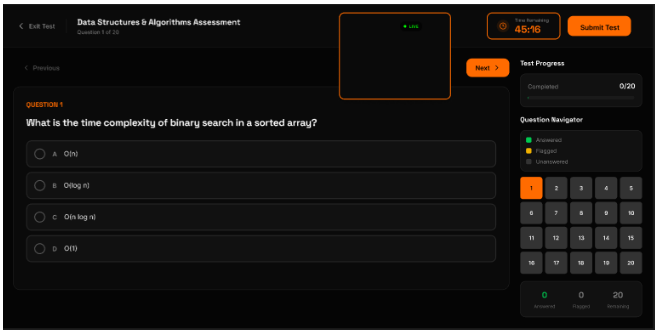

Figure 4.4. The Quiz Interface for evaluating user knowledge.

During this session, the Flask-based AI engine processes the webcam feed in real-time.

1. **Normal State (Focused):** In Figure 4.5, the user is looking directly at the screen with eyes open. The system calculates the Eye Aspect Ratio (EAR) > 0.25 and Head Pose rotation $< 20^{\circ}$. Consequently, the status is correctly displayed as **"Fokus"** (Green Indicator).

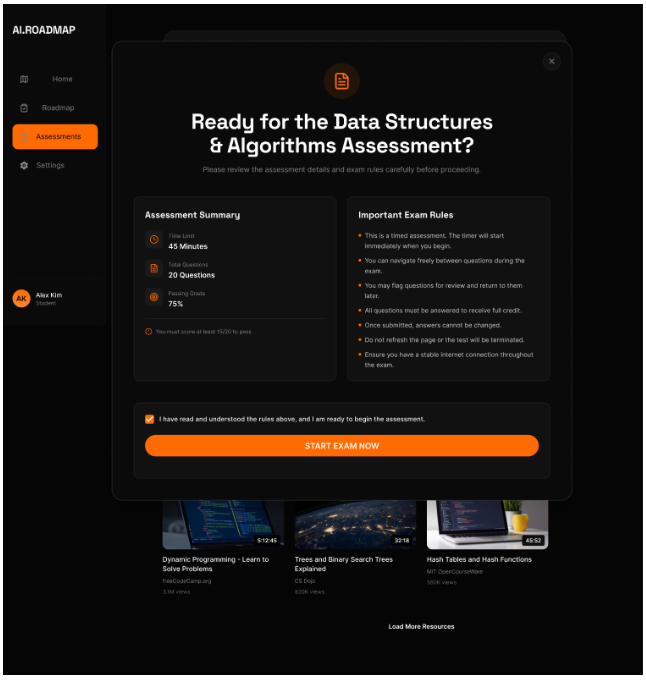

Figure 4.5. Active Monitoring showing the user is Focused.

2. **Anomaly Detection (Distracted/Drowsy):** To evaluate the system's sensitivity, we simulated a "cheating" or "sleeping" behavior. As captured in **Figure 4.6**, when the user closes their eyes or looks away from the screen, the EAR drops below the threshold. The system successfully detects this anomaly and triggers an immediate alert: **"Peringatan! Tidak Fokus / Mata Tertutup"** (Warning! Not Focused / Eyes Closed).

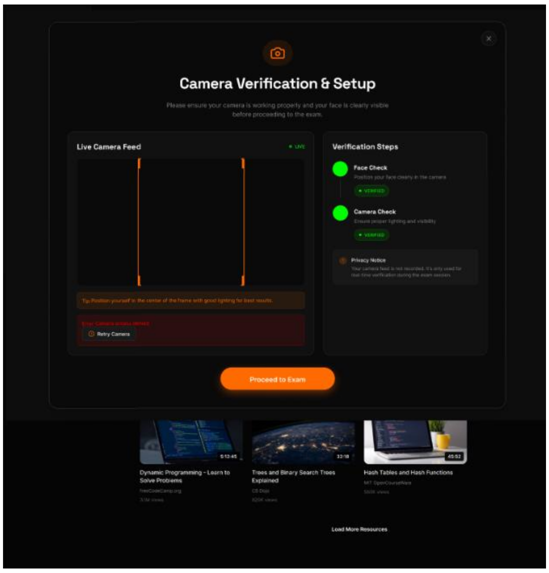

Figure 4.6. Anomaly Detection: System warns of distraction/sleeping.

This experiment proves that the integration between the Next.js frontend (displaying the alert) and the Python Flask backend (processing the video frames) operates seamlessly with low latency.

C. Assessment Completion

Upon completing the quiz without violating the integrity constraints, the user receives their final score. Figure 4.7 shows the result page, completing the educational loop.

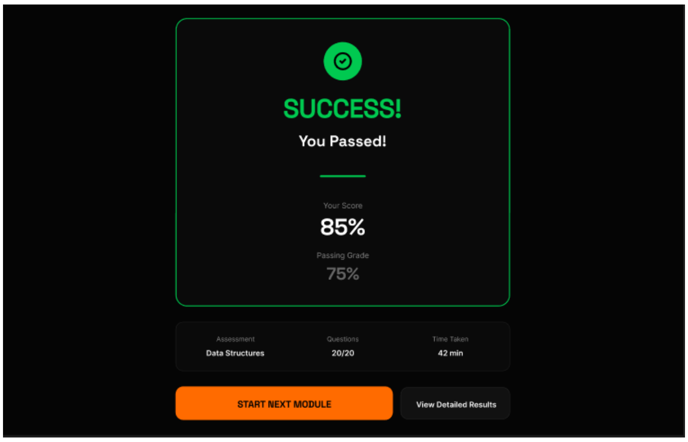

Figure 4.7. Final Assessment Res9

III. Conclusion and Future Works

This project successfully designed and implemented "AI Roadmap," a web-based educational platform developed to address the critical challenges of unstructured learning paths and the lack of supervision in online education.

The development process demonstrated that the **Decision Support System (DSS)**, utilizing rule-based filtering, was highly effective in solving the confusion often faced by beginners. By analyzing user inputs specifically their role, focus area, and goals the system successfully generated structured and relevant learning paths, differentiating clearly between the needs of a Data Scientist, a Computer Vision Engineer, and other specializations.

Furthermore, the integration of **Real-Time Face Recognition** proved to be a robust solution for automated proctoring. The system demonstrated high sensitivity in detecting behavioral anomalies, utilizing the Eye Aspect Ratio (EAR) and Head Pose Estimation to identify drowsiness and distraction. This ensures that the platform not only recommends *what* to learn but also enforces *how* the user learns with focus and integrity. Finally, the seamless integration between the **Next.js** frontend and the **Python Flask** backend confirmed that a decoupled architecture is a scalable and viable solution for deploying complex Deep Learning models on the web.

In summary, this platform provides a holistic technical solution that fosters a more focused, personalized, and honest digital learning environment.

IV. References

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